

Precalculus

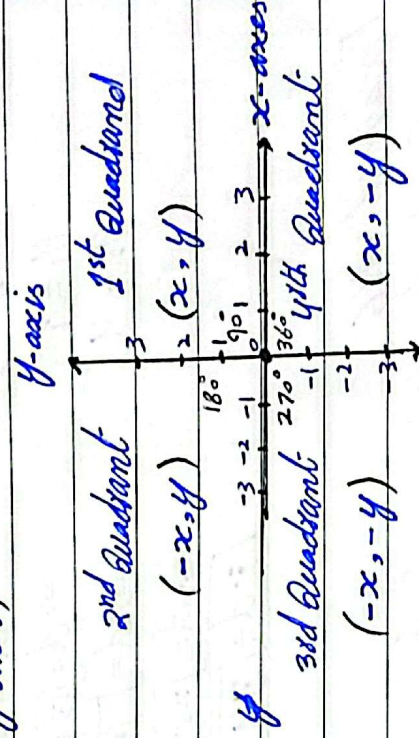
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Rectangular Coordinates

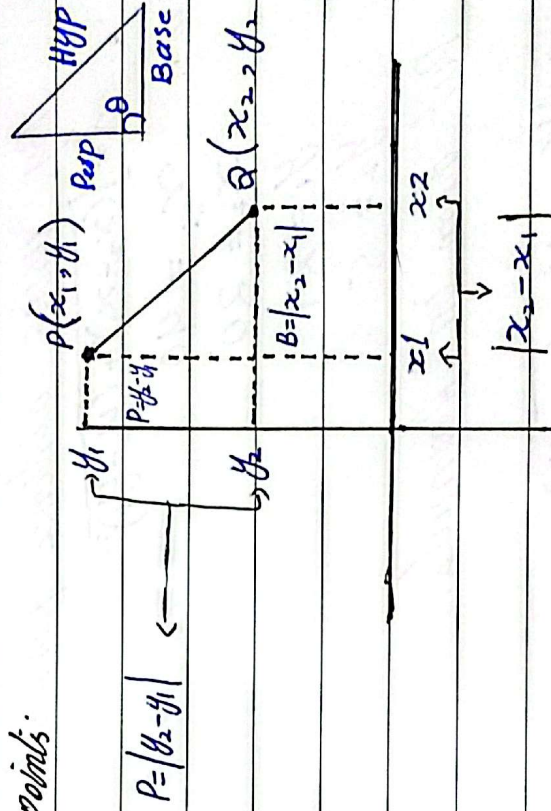
∴ Also known as Cartesian coordinate.

⇒ It is two dimensional system for locating points using two perpendicular lines, such as the horizontal line (x-axis) and vertical line (y-axis)



Pythagorean Theorem and Distance Formula

→ Distance between two points



According to Pythagorean Theorem: $H^2 = B^2 + P^2$

$$\Rightarrow d^2 = |x_2 - x_1|^2 + |y_2 - y_1|^2$$

$$\Rightarrow d = \sqrt{|x_2 - x_1|^2 + |y_2 - y_1|^2}$$

$$\therefore d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Formula.

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Example #3

Find distance between two points: $(-2, 1)$ and $(3, 4)$

By using distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Here: $x_1 = -2$

$x_2 = 3$

$y_1 = 1$

$y_2 = 4$

$\Rightarrow$ Solution

$$\therefore d = \sqrt{(3 - (-2))^2 + (4 - 1)^2}$$

$$\Rightarrow d = \sqrt{(3 + 2)^2 + (4 - 1)^2}$$

$$\Rightarrow d = \sqrt{(5)^2 + (3)^2}$$

$$\Rightarrow d = \sqrt{25 + 9}$$

$$\therefore d = \sqrt{34}$$

Example: #6

Find the distance between two points:

$P(40, 29)$ and $Q(20, 5)$

By using distance formula.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Here: $x_1 = 40$

$x_2 = 20$

$y_1 = 29$

$y_2 = 5$

Put them in formula.

$$\Rightarrow d = \sqrt{(20 - 40)^2 + (5 - 29)^2}$$

$$\Rightarrow d = \sqrt{(-20)^2 + (-24)^2}$$

$$\Rightarrow d = \sqrt{400 + 576}$$

$$\boxed{d = \sqrt{976}} \text{ Ans.}$$

Simple definition: The midpoint formula finds the middle point a line segment between two points.

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The midpoint Formula

To find the midpoint of the line segment that joins two points in a coordinate plane, find the average values of the respective coordinates of the two endpoints. Using the formula Midpoint formula.

Midpoint formula

$$\therefore x - x_1 = x_2 - x$$

$$\Rightarrow x + x = x_2 + x_1 \therefore \text{By rearranging}$$

$$\Rightarrow 2x = x_1 + x_2$$

$$\therefore x = \frac{x_1 + x_2}{2}$$

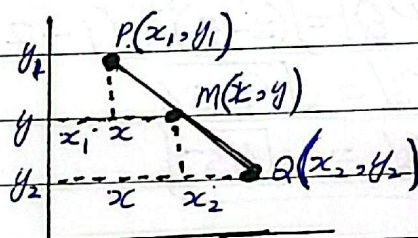
$$\therefore y - y_1 = y_2 - y$$

$$\Rightarrow y + y = y_2 + y_1$$

$$\Rightarrow 2y = y_1 + y_2$$

$$\therefore y = \frac{y_1 + y_2}{2}$$

Note Both side
of midpoint
should be
equal



$$\left. \begin{aligned} |x - x_1| &= |x_2 - x| \\ |y - y_1| &= |y_2 - y| \end{aligned} \right\} \rightarrow \text{Remove modulus}$$

Example # 6

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* Show that the points of vertex of $(2,1)$, $(4,0)$ and $(5,7)$ are right angle triangle.

According to Pythagorean Theorem:

$$\therefore H^2 = B^2 + P^2$$

$$\therefore |PQ| = \sqrt{(2-5)^2 + (1-7)^2}$$

$$\Rightarrow |PQ| = \sqrt{9+36}$$

$$\therefore |PQ| = \sqrt{45}$$

$$\therefore |PR| = \sqrt{(4-5)^2 + (0-7)^2}$$

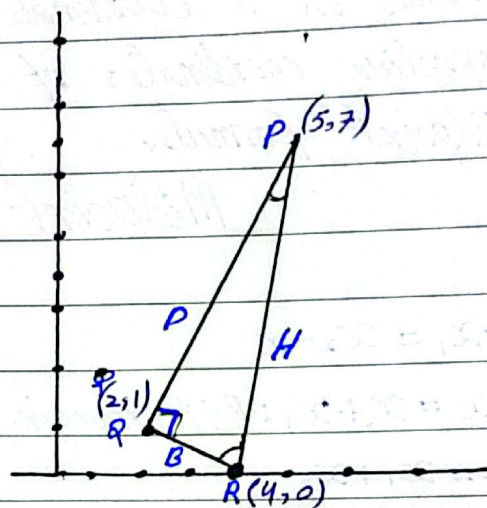
$$\Rightarrow |PR| = \sqrt{1+49}$$

$$\therefore |PR| = \sqrt{50}$$

$$\therefore |QR| = \sqrt{(4-2)^2 + (1-0)^2}$$

$$\Rightarrow |QR| = \sqrt{(2)^2 + (1)^2}$$

$$\therefore |QR| = \sqrt{5}$$



Now put the value of PQ, PR, QR in Pythagorean Theorem.

$$|QR|^2 = |PQ|^2 + |PR|^2$$

$$\Rightarrow |PR| = |PQ| + |QR|$$

$$\Rightarrow \sqrt{50} = (\sqrt{45})^2 + (\sqrt{5})^2$$

$$\Rightarrow 50 = 45 + 5$$

$$\therefore 50 = 50 \text{ Hence proved.}$$

Equations and Graphs

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* What is equation:

∴ A mathematical statement that indicates two mathematical equations/expressions are equal.

It uses an equal sign $=$ to show that the expression on other side of ~~sig~~ sign have the same value.

example: $2x - 5 = 13$

$$5x^2 + 2x - 8 = 16$$

Graph of linear equation:

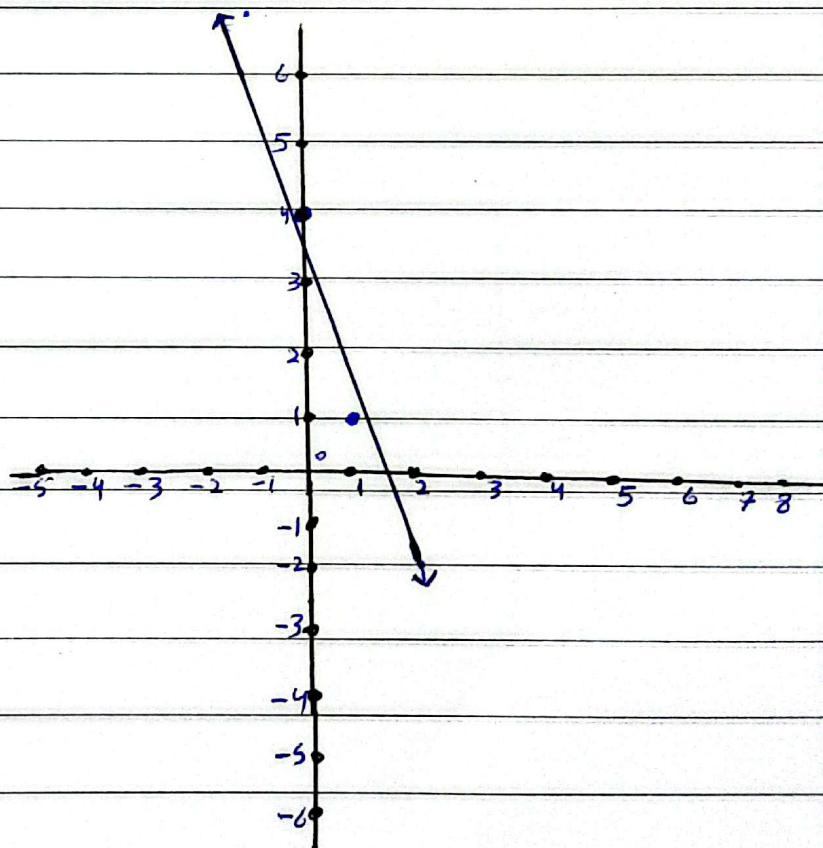
∴ Graph of linear equation is always straight line.

example:

Draw the graph of linear equation

* $3x + y = 4$

x	y
0	4
1	1
-1	7
2	-2
-2	10
3	-5
-3	13



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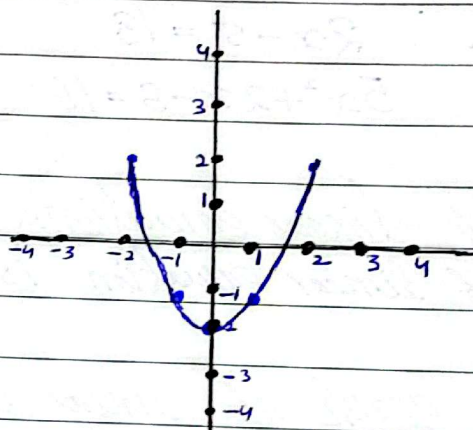
Graph of Quadratic equation

Quadratic equation:-

A mathematical equation where the highest power of variable is two, written in the standard form $ax^2 + bx + c = 0$.

* $y = x^2 - 2$

x	y
0	-2
1	-1
-1	-1
2	2
-2	2



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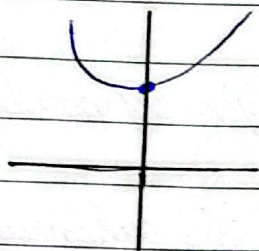
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The intercept of a graph.

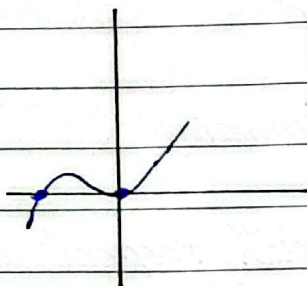
∴ A point where a line or curve crosses an axis on a coordinate plane.

- The x-intercept is the point where the graph crosses the x-axis, and at that point, the y coordinate is always zero.
- The y-intercept is the point where the graph crosses the y-axis, and at that point x-coordinate is always zero.

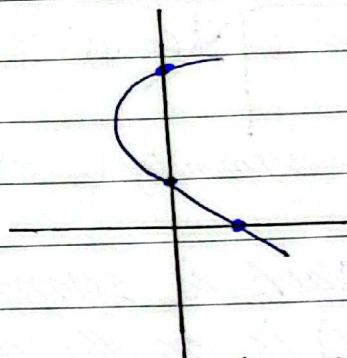
Examples:



- No x intercept
- one y-intercept



- 2 x-intercept
- one y-intercept



- one x-intercept
- 2 y-intercept

Example: Find the x and y intercept of eq

$$x + 2y = 3$$

∴ For x-intercept put $y = 0$

$$= x + 2(0) = 3$$

$$= x + 0 = 3$$

$$\boxed{\therefore x = 3} \Rightarrow (3, 0)$$

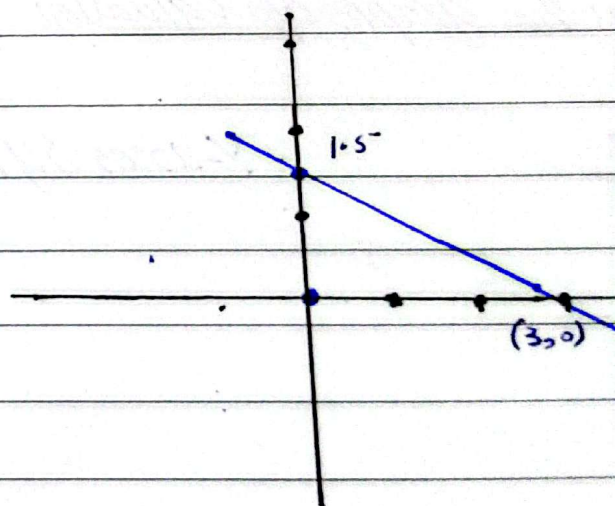
∴ For y-intercept put $x = 0$

$$\therefore x + 2y = 3$$

$$\Rightarrow 0 + 2y = 3$$

$$\Rightarrow y = \frac{3}{2}$$

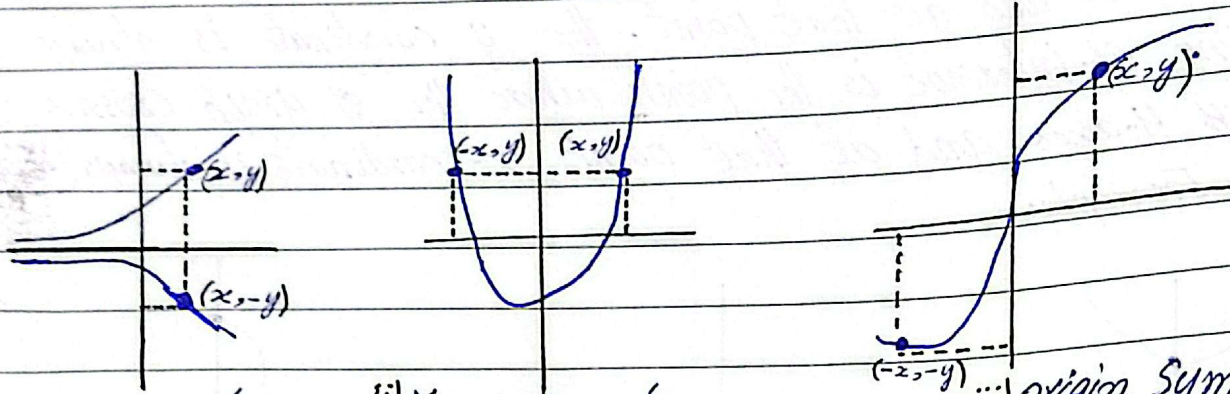
$$\boxed{\therefore y = 1.5}$$



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Symmetry:

Graphs of equation can have symmetry with respect to one of the coordinate axes or with respect to the origin.



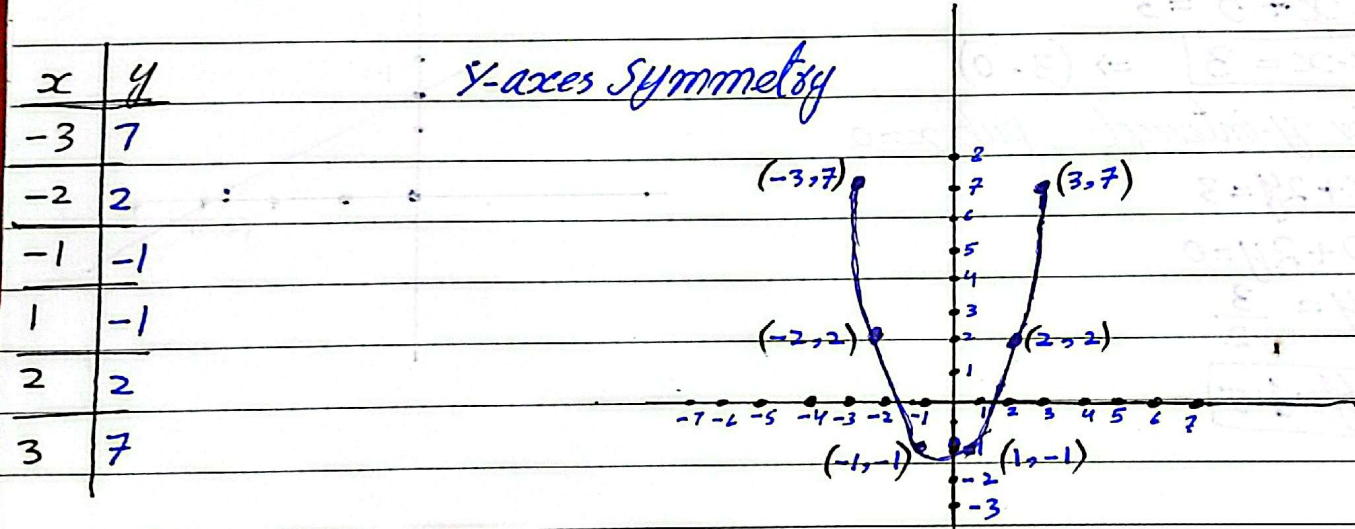
i) x-axis symmetry

ii) y-axis symmetry

iii) origin symmetry

- i) A graph is symmetric with respect to x-axis if, whenever (x, y) is on the graph, $(x, -y)$ is also on the graph.
- ii) A graph is symmetric with respect to y-axis if, whenever (x, y) is on the graph, $(-x, y)$ is also on the graph.
- iii) A graph is symmetric with respect to origin if, whenever (x, y) is on the graph, $(-x, -y)$ is also on the graph.

Draw the graph of equation $y = x^2 - 2$ with respect to y-axis.



- 1) The graph of equation is symmetric with respect to x-axes when y with $-y$ gives same equation.

e.g. $y = 2x^3$

$$(-y) = 2x^3$$

$$\boxed{-y = 2x^3}$$

$\therefore$ By replacing y with $-y$ w.r.t x-axes
 $\therefore$ Equation is not symmetric w.r.t x-axes.

- 2) The graph of equation is symmetric with respect to y-axes, when replacing x with $-x$, gives same equation.

e.g. $y = 2x^3$

$$y = 2(-x)^3$$

$$\boxed{y = -2x^3}$$

$\therefore$ By replacing x with $-x$ w.r.t y-axes

$\therefore$ Equation is not symmetric with respect to y-axes.

- 3) The graph of equation is symmetric with respect to origin when replacing x with $-x$, and y with $-y$ gives same equation.

e.g. $y = 2x^3$

$$-y = 2(-x)^3$$

$$-y = -2x^3$$

$$\boxed{y = 2x^3}$$

$\therefore$ By replacing x with $-x$ and y with $-y$ with respect to origin.

$\therefore$ Equation is symmetric with respect to origin.

Date: / / Linear equations in two variables: M T W T F S S

Slope :: Tangential ratio of inclination
OR: It is the ratio of rise and run.
OR: Steepness of line.

$$\therefore m = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{Formula of slope.}$$

Example :: Find the slope between (2, 0) and (1, 4) the point.

$$\begin{aligned} \therefore m &= \frac{y_2 - y_1}{x_2 - x_1} & \text{Here: } x_2 &= 1, & y_2 &= 4 \\ & & x_1 &= 2, & y_1 &= 0 \\ m &= \frac{4 - 0}{1 - 2} = \frac{4}{-1} = -4 \text{ Ans.} \end{aligned}$$

* Slope intercept form of the equation of a line:

$$\therefore y = mx + b$$

Slope = m y-intercept = b

Example :: $5x + 10y = 0$

$$\therefore 10y = -5x$$

$$\Rightarrow y = -\frac{5}{10}x$$

$$\Rightarrow y = -\frac{1}{2}x$$

$$\therefore m = -\frac{1}{2} \rightarrow \text{y-intercept} = 0$$

Ans

Example :: $6x + 2y = 7$

$$2y = 7 - 6x$$

$$y = -\frac{6}{2}x + \frac{7}{2}$$

$$y = -3x + \frac{7}{2} \text{ Ans.}$$

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* Point Slope formula:

$$\Rightarrow (y - y_1) = m(x - x_1)$$

Example::

$$m = 5 \quad P(x, y) = (2, 3)$$

By putting values on formula.

$$\Rightarrow y - 3 = 5(x - 2)$$

$$= y - 3 = 5x - 10$$

$$= y = 5x - 10 + 3$$

$$\therefore y = 5x - 7 \text{ Ans}$$

Example::

Find the slope intercept form of the equation that has a slope of 3 and passes through the point: (1, -2)

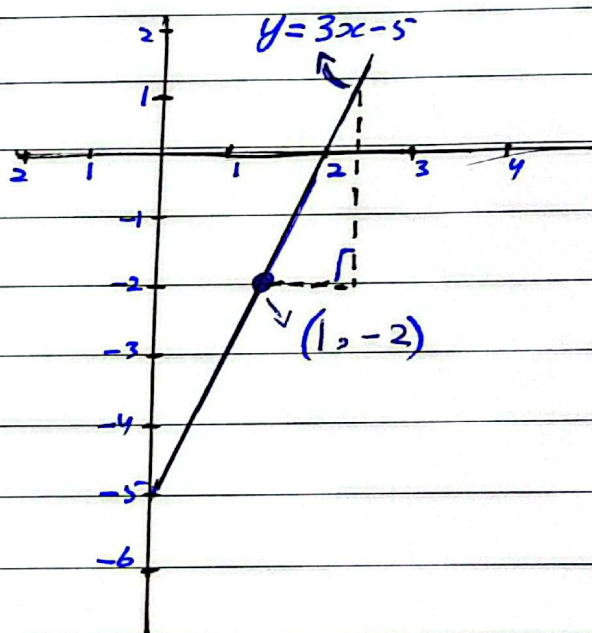
$$\therefore y - y_1 = m(x - x_1)$$

$$\Rightarrow y - (-2) = 3(x - 1)$$

$$\Rightarrow y + 2 = 3x - 3$$

$$\Rightarrow y = 3x - 3 - 2$$

$$\therefore y = 3x - 5 \text{ Ans}$$

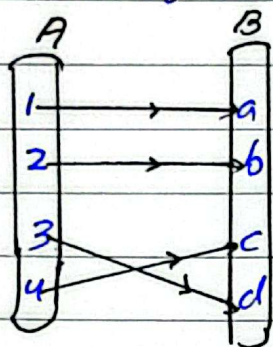


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Functions

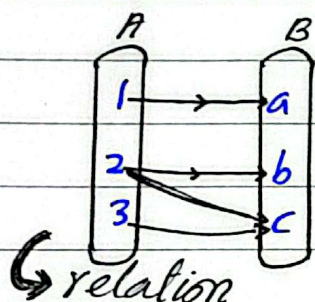
: A function $F: A \rightarrow B$ is a relationship between two variables such that to each value of independent value their corresponds only one value of the dependent variable.



- All element of set A must make relation with one of set B. No left over in set A
- If B set element can be left over.

* Relation: It is set of ordered pair or sub set of $A \times B$ of two sets.

$\{(a, b) (c, a) (2, b) (2, c) (3, a)\}$



relation

Characteristics of function:

- * Each element in A must be matched with an element of B.
- * Some element in B may not be matched with any element in A.
- * Two or more elements in A maybe matched with some element in B.
- * An element in B can not be matched with two different element in A.

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∴ Domain: The domain of a function is the set of all values inputs of independent variables for which the function is defined.

∴ Range: The range of function is the set of all values output taken by the dependent variable.

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Example #02:

Determine, whether the equation represents the function or not.

i) $y^2 + x = 1$

Solution: To determine whether y is a function of x , solve for y in terms of x .

$$x^2 + y = 1$$

$$\therefore y = 1 - x^2$$

$\Rightarrow$ To each value of x there corresponds exactly one value of y , so y is function of x

ii) $-x + y^2 = 1$

$$y^2 = 1 + x$$

$$y = \pm \sqrt{x+1}$$

$$y = \pm \sqrt{1+x}$$

The $\pm$ indicates that to a given value of x there is correspond two values of y so, y is not a function of x .

[If power of y is even = not function (eg: $x = y^2 / y^2 = x$)
If power of y is odd = function (eg: $y^3 + x = 1, y = x, y^3 = x+1$)]

$\therefore y^3 = x$ can be written as $y = \sqrt[3]{x}$, which is function because for each value/input x there is one output y .

$\Rightarrow$ If $x=8$, then $y = \sqrt[3]{8} = 2$, there is only one y value for $x=8$.

Example: Determine whether is function or not.

i) $x^2 + y^2 = 8$

$$x^2 + y^2 = 8$$

$$y^2 = 8 - x^2$$

$$y = \pm \sqrt{8 - x^2}$$

Not a function

ii) $y - 4x^2 = 36$

$$y - 4x^2 = 36$$

$$y = 36 + 4(1)^2$$

$$y = 36 + 4$$

$$y = 40$$

It function.

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Evaluating a Function:

Let $g(x) = -x^2 + 4x + 1$. Find each function value.

a) $g(2)$

$g(x) = -x^2 + 4x + 1$

replace x with 2

$g(2) = -(2)^2 + 4(2) + 1$

$g(2) = -4 + 8 + 1$

$\therefore g(2) = 5$

b) $g(t)$

$g(x) = -x^2 + 4x + 1$

replace x with t

$g(t) = -(t)^2 + 4t + 1$

$g(t) = -t^2 + 4t + 1$

$g(t) = -t^2 + 4t + 1$

c) $g(x+2)$

$g(x) = -x^2 + 4x + 1$

replace x with $x+2$

$g(x+2) = -(x+2)^2 + 4(x+2) + 1$

$g(x+2) = -x^2 + 2(x)(2) + 2^2 + 4x + 8 + 1$

$g(x+2) = -x^2 + 4x + 4 + 4x + 8 + 1$

$g(x+2) = -x^2 + 5$

* Piecewise function:

In math problems, one single rule is not enough for all values of x . So we split the domain into pieces and give a different formula for each piece. So that's why it's called piecewise.

⇒ Proper definition:

A piecewise function is a function defined by multiple rules, each applying to a specific interval of the output.

$$f(x) = \begin{cases} x^2 + 1 & \text{if } x < 0 \\ x + 1 & \text{if } x \geq 0 \end{cases}$$

• For $x = -2$ we use x^2 so,

$\Rightarrow f(-2) = (-2)^2 = 4$

• for $x = 3$ we use $x + 1$, so

$\Rightarrow f(3) = 3 + 1 = 4$

Example:

Find the all values of x for which $f(x) = 0$

a) $f(x) = -2x + 10$

$= -2x + 10 = 0$

$x = -2x = -10$

$= x = \frac{-10}{-2} = 5 \text{ Ans.}$

So $f(x) = 0$ when $x = 5$

Prove it: $f(x) = 5$

$-2(5) + 10$

$f(x) = -10 + 10 = 0$

b) $f(x) = x^2 - 5x + 6$

$x^2 - 5x + 6 = 0$

$x^2 - 2x - 3x + 6$

$x(x-2) - 3(x-2)$

$(x-3)(x-2)$

$x = 3, x = 2$

So $f(x) = 0$ when $x = 2$ or $x = 3$

Prove it

when $f(x) = 2 = f(2) = (2)^2 - 5(2) + 6$

$\Rightarrow f(2) = 4 - 10 + 6 = 0$

when $f(x) = 3 = f(3) = (3)^2 - 5(3) + 6$

$\Rightarrow f(3) = 9 - 15 + 6 = 0$

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Find the value for which $f(x) = g(x)$

→ Find the values of x for which $f(x) = g(x)$

a) $f(x) = x^2 + 1$ and $g(x) = 3x - x^2$

Solution:

$$f(x) = g(x)$$

$$x^2 + 1 = 3x - x^2$$

$$2x^2 - 3x + 1 = 0 \Rightarrow \text{General form}$$

$$2x^2 - 2x - x + 1 = 0 \Rightarrow \text{Factor}$$

$$2x(x-1) - 1(x-1) = 0$$

$$(2x-1)(x-1) = 0$$

$$\Rightarrow 2x-1=0 \Rightarrow x = \frac{1}{2}$$

$$\Rightarrow x-1=0 \Rightarrow x = 1$$

So $f(x) = g(x)$ when: $x = \frac{1}{2}$ or $x = 1$

$\Rightarrow$ If we put value of $x = \frac{1}{2}$ or $x = 1$ Then we will come to know that $f(x) = g(x)$

Prove it:

$$\therefore f(x) = x^2 + 1, \quad g(x) = 3x - x^2$$

$$\Rightarrow f\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 + 1, \quad g\left(\frac{1}{2}\right) = 3\left(\frac{1}{2}\right) - \left(\frac{1}{2}\right)^2$$

$$f\left(\frac{1}{2}\right) = \frac{1}{4} + 1, \quad g\left(\frac{1}{2}\right) = \frac{3}{2} - \frac{1}{4}$$

$$f\left(\frac{1}{2}\right) = \frac{1+4}{4}, \quad g\left(\frac{1}{2}\right) = \frac{6-1}{4} \text{ By LCM}$$

$$f\left(\frac{1}{2}\right) = \frac{5}{4}, \quad g\left(\frac{1}{2}\right) = \frac{5}{4}$$

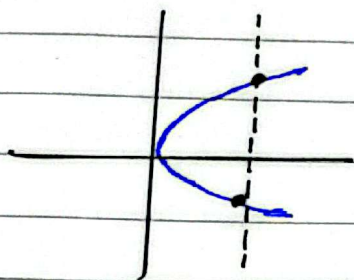
Hence proved $f(x) = g(x)$

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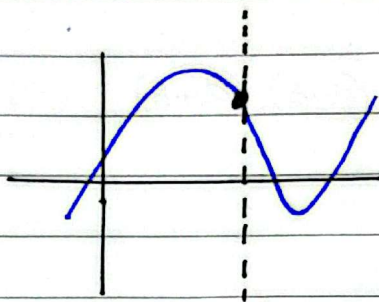
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Vertical line test:

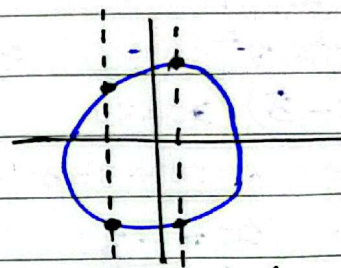
It can be used to determine whether a graph represents a function. If we draw a vertical line that intersects a graph once along each point, then it is a function. If vertical line intersects more than once, then graph does not define a function.



* Not a function



* A function



* Not a function.

Zero of function

The zero of a function is the value of x for which the function equals zero. ($f(x)=0$)

Example:

a) $f(x) = 3x^2 + x - 10$

$$f(x) = 3x^2 + x - 10$$

$$= (3x - 5)(x + 2) = 0$$

$$= 3x = 5, \quad x = -2$$

$$\boxed{x = \frac{5}{3}, \quad x = -2}$$

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Zero of the function.

∴ If the graph of a function of x has an x intercept at $(a, 0)$, then a is zero of the function.

OR ∴ The zero of a function is the value of x that makes function equal to zero.

Example:

$$f(x) = x^2 - 9$$

$$f(x) = x^2 - 9$$

$$= x^2 - 3^2$$

$$\therefore a^2 - b^2 = (a - b)(a + b)$$

$$\text{So, } (x - 3)(x + 3)$$

$$\therefore x - 3 = 0, \quad x + 3 = 0$$

$$\therefore \underline{x = 3}, \quad \underline{x = -3}$$

Ans Ans

If we put the final answer then function will become zero;

$$x = 3$$

$$x = -3$$

$$\Rightarrow f(x) = x^2 - 9$$

$$\Rightarrow f(x) = x^2 - 9$$

$$\Rightarrow f(3) = 3^2 - 9$$

$$\Rightarrow f(-3) = (-3)^2 - 9$$

$$\Rightarrow f(x) = 9 - 9$$

$$\Rightarrow f(3) = 9 - 9$$

$$\Rightarrow f(x) = 0$$

$$\Rightarrow f(3) = 0$$

So these both values makes x the function zero.

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Even and Odd functions.

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In the terminology of function, a function is said to be:

- even: when its graph is symmetric w.r.t y-axis.
- odd: when its graph is symmetric w.r.t the origin.

→ Even function:

∴ A function is even if: $f(-x) = f(x)$

→ This means the graph is with respect to y-axis.

example:

∴ Quick test →

$$1) f(x) = x^2$$

$$\Rightarrow f(-x) = (-x)^2$$

$$= x^2 \text{ Ans.}$$

$$2) g(x) = x^4 - x^2 - 1$$

$$\Rightarrow g(x) = x^4 - x^2 - 1$$

• Replace x with $-x$ ∴ $f(-x) = f(x)$

$$\Rightarrow g(-x) = (-x)^4 - (-x)^2 - 1$$

$$\Rightarrow \boxed{g(-x) = x^4 - x^2 - 1} \text{ Ans.}$$

Note:

• If power of x are all even → even function.

• If power of x are all odd → odd function.

• If mixture = neither even nor odd.

○ → Odd function:

∴ A function is odd if: $f(-x) = -f(x)$

Example:

$$1) f(x) = x^3$$

$$f(x) = x^3 = f(-x) = (-x)^3 = -x^3 = -f(x)$$

$$2) f(x) = 2x^3 + 3x$$

$$\Rightarrow f(-x) = -f(x)$$

$$\Rightarrow f(-x) = -2x^3 + 3x$$

$$\Rightarrow f(-x) = -(2x^3 + 3x)$$

$$\Rightarrow \boxed{f(-x) = -2x^3 + 3x}$$

Neither even nor odd function.

∴ If the function has a mixture of powers.

Example:

$$\Rightarrow f(x) = x^2 + x \quad \phi$$

$$\Rightarrow f(-x) = (-x)^2 + (-x) = x^2 - x$$

$$\Rightarrow f(-x) = x^2 - x \text{ Ans.}$$

Neither even nor odd function.

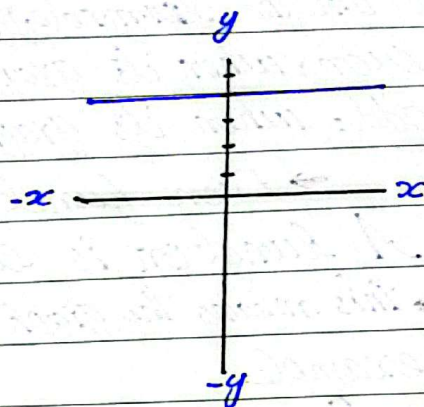
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∴ Constant function: $f(x) = c$

A constant function is a function that always gives the same output, no matter what x is

→ $f(x) = 4$



Graph: The graph is horizontal line parallel to the x -axis, crossing the y -axis at $(0, c)$

Point to be noted that: 😊 For constant function of the form $f(x) = c$

- The value of $f(x)$ (or y) is always c .
- It does not depend on the input x .
- That's why we say there is no independent variable affecting the output.

So:

Linear function → depends on x .
Quadratic function → depends on x .
Constant function → Independent of x .

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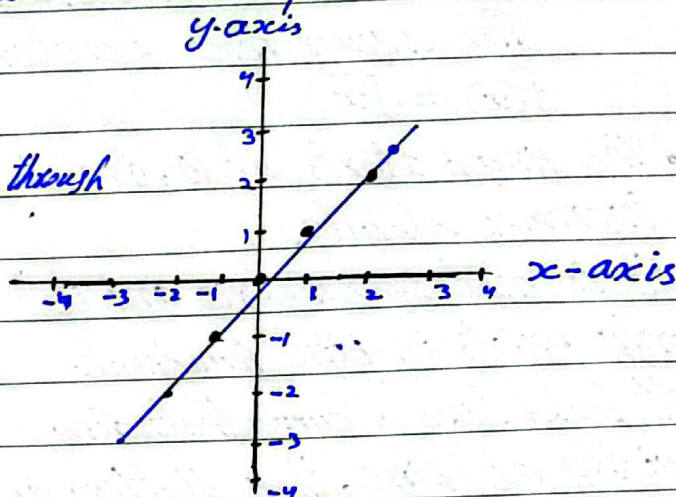
∴ Identity function:-

It is defined as $f(x) = x$

It simply returns the same value at the input.

x	y
-2	-2
-1	-1
0	0
1	1
2	2

- For any input value of x , the output will be the same.
- Graph is straight line passing through the origin $(0,0)$.
- It makes a 45° angle with both x -axis and y -axis.



∴ Absolute value function.

The absolute value of number is its distance from 0 on the number line, always a non-negative number.

It is written as:-

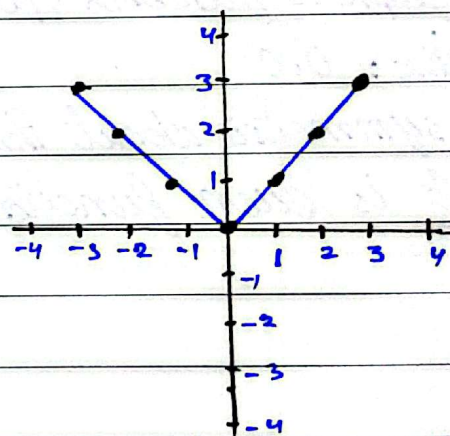
$$f(x) = |x|$$

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* The graph of absolute value function is V shaped.

* It has vertex at $(0,0)$ and is symmetric with y -axis so it's an even function.

x	y
-3	3
-2	2
-1	1
0	0
1	1
2	2
3	3



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∴ Square root function:

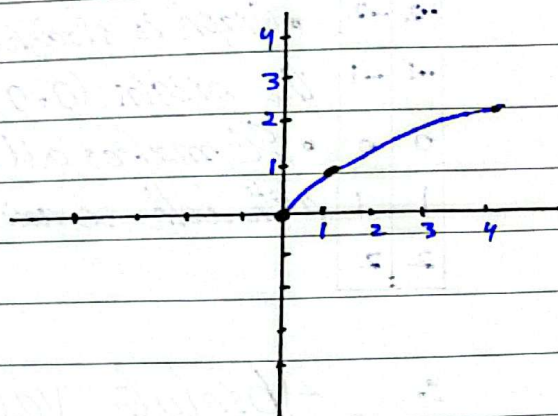
It takes non negative input x and gives the non-negative number whose square is x .

The square root function is:

$$f(x) = \sqrt{x}$$

- The graph starts at $(0,0)$ and increases slowly.
- It is not symmetric and it only exists in first quadrant.
- It looks like a curve that rises quickly at first, then flattens as x gets larger.
- Always goes up as x increases.
- Not even nor odd.

$\sqrt{x}$	y
$\sqrt{0}$	0
$\sqrt{1}$	1
$\sqrt{4}$	2
$\sqrt{9}$	3
$\sqrt{16}$	4



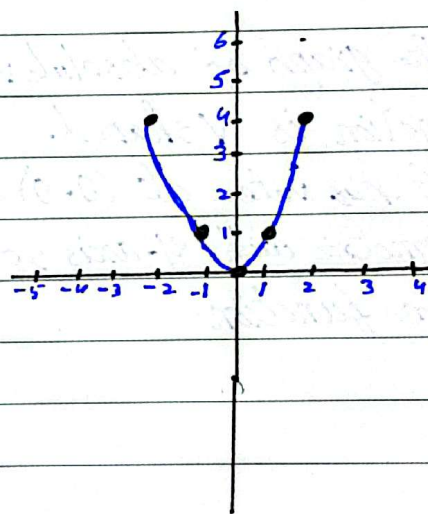
∴ Squaring function:

It takes a number x and multiplies by itself also we can say that the highest power is 2.

$$f(x) = x^2$$

- The graph is parabola opening upward.
- Vertex: $0,0$
- It is symmetric about the y -axis,
- so it is even function.

x	y
-2	4
-1	1
0	0
1	1
2	4



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∴ Cubic function.

It is the type of a polynomial function with highest power 3.

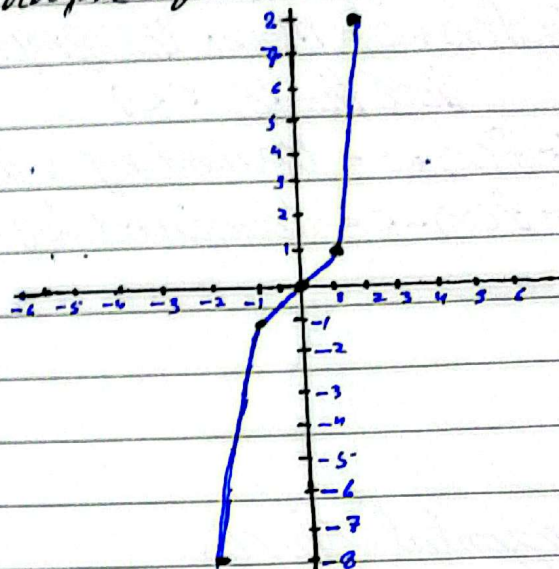
Its general form is:

$$f(x) = ax^3 + bx^2 + cx + d$$

Graph of $f(x) = x^3$

- It is symmetric w.r.t origin.
- If $a > 0$: goes down left, up right.
- If $a < 0$: goes up left, down right.

x	y
-2	-8
-1	-1
0	0
1	1
2	8



∴ Reciprocal function.

A reciprocal function is a function that contains a variable in the denominator.

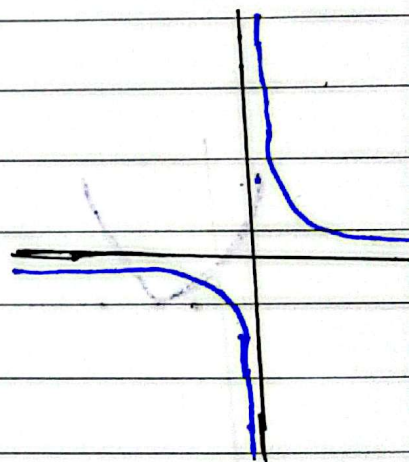
General form:

$$f(x) = \frac{1}{x}$$

- The function is undefined at $x=0$.
- Domain: $(-\infty, 0) \cup (0, \infty)$
- The function can take all real values except 0.
- No x-intercept and y-intercept.
- decreasing on $(0, \infty)$
- increasing on $(-\infty, 0)$

x	y
-2	-0.5
-1	-1
-0.5	-2
0.5	2
1	1
2	0.5

y-axis



• Two branches:

- Quadrant 1: positive x and y.
- Quadrant 2: negative x and y.

∴ Transformation function.

A transformation function changes the graph of a given function in some way.

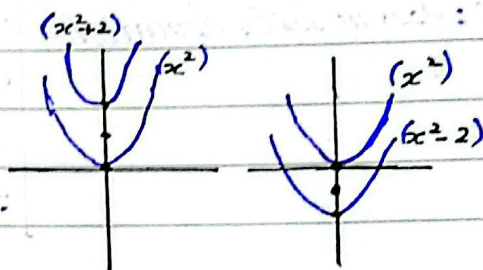
Let's say we have a base function $f(x)$

a) Vertical shift ∴

• $y = f(x) + c$ = moves the graph up by c units if $c > 0$ or down if $c < 0$

⇒ $y = f(x) + c$ = upward by c units

⇒ $y = f(x) - c$ = downward by $-c$ units.



i) $f(x) = x^2 + 2$

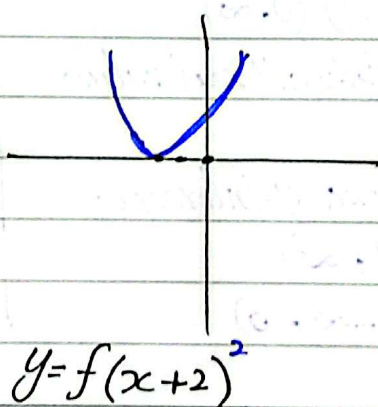
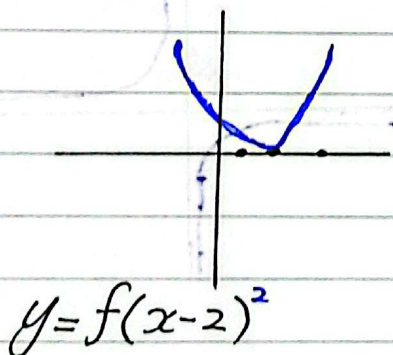
$f(x) = x^2 - 2$

b) Horizontal shift.

$y = f(x+h)$ = shift the graph to left side if $c < 0$ and if $c > 0$ then moves towards right side.

• $y = f(x-2)$ = right 2 unit

• $y = f(x+2)$ = left 2 unit.



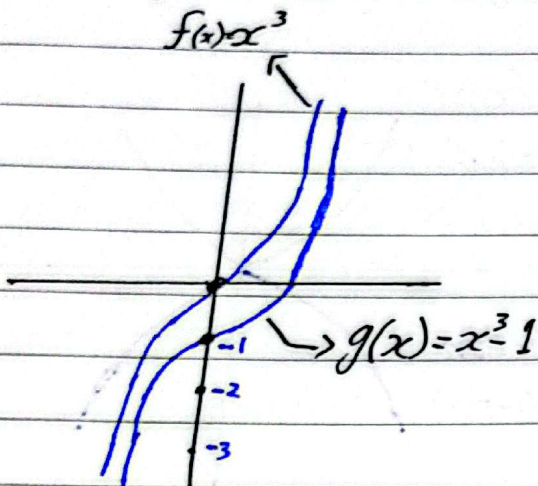
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Example: Shifting the graph of function:

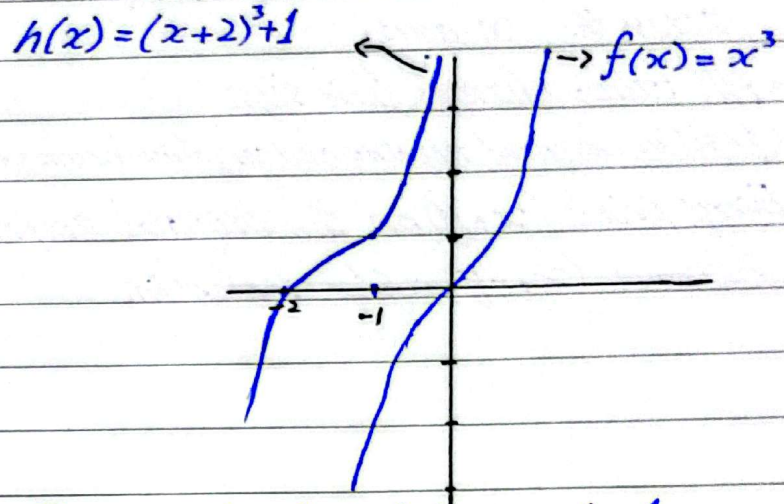
use the graph of $f(x) = x^3$ to sketch the graph each function.

a) $g(x) = x^3 - 1$



This graph is shifted downward by one unit.

b) $h(x) = (x+2)^3 + 1$



This graph shifted left side by 2 unit and shifted upward by 1 unit.

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∴ Reflection of graph:

Reflection of a graph means flipping the graph of the function across a line, usually the x-axis or y-axis.

⇒ Reflection across the x-axis:

- If " $y=f(x)$ " is the original function, then " $y=-f(x)$ " is the reflection across the x-axis.

All values change their sign

(positive becomes negative and negative becomes +ve)

Every point (x, y) on the original graph

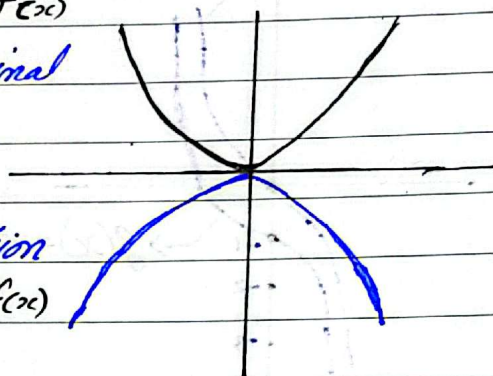
becomes $(x, -y)$ after reflection

$$y = f(x)$$

• original

• Reflection

$$y = -f(x)$$



⇒ Reflection across the y-axis:

If $y=f(x)$ is the original function, then $y=f(-x)$ is the reflection across the y-axis.

$\therefore$ Reflection and Shift

Shift:

Shifting a graph means moving it without changing its shape. There are two kinds:

i) Horizontal Shift:

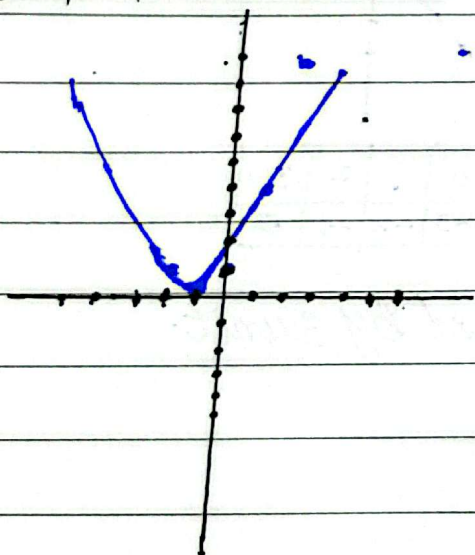
A type of transformation in which entire graph of a function $y=f(x)$ is moved left or right on the coordinate plane without changing its shape, size or orientation.

- $y=f(x+c)$ = shift of the $y=f(x)$ c units to the left.
- $y=f(x-c)$ = shift of the $y=f(x)$ c unit to the right.

1) $y=x^2$

a) $y=(x+1)^2$

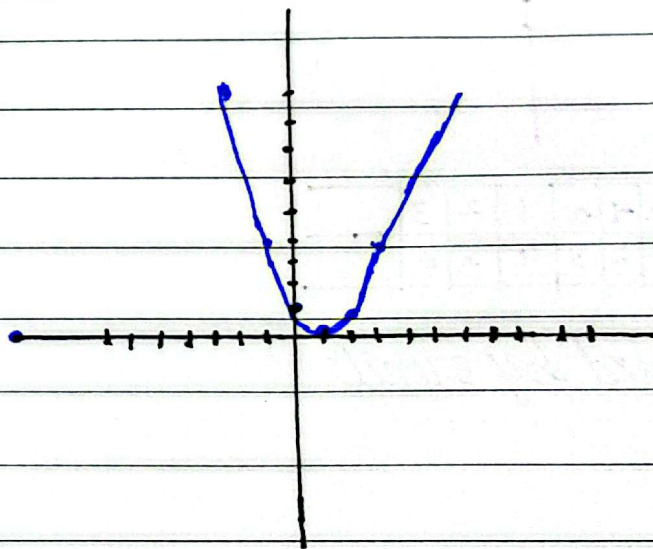
x	-2	-1	0	1	2
y	1	0	1	4	9



Shifted left side by 1

b) $y=(x-1)^2$

x	-2	-1	0	1	2	3
y	9	4	1	0	1	4



Shifted right side by one.

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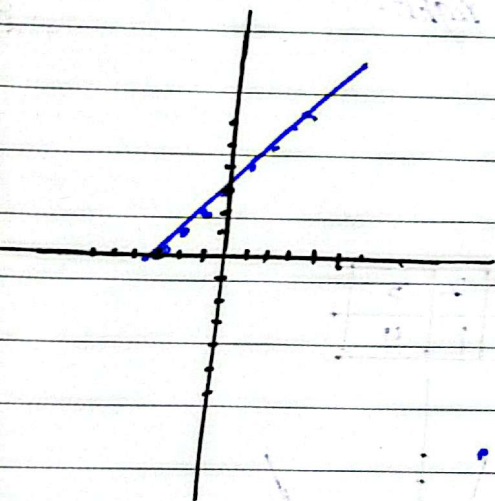
iv) Vertical Shift:

A vertical shift moves the graph of a function up or down on the coordinate plane without changing its shape or orientation.

- $y = x + c$: Shift of $y = f(x)$ c unit upward.
- $y = x - c$: Shift of $y = f(x)$ c unit down.

1. $y = x$

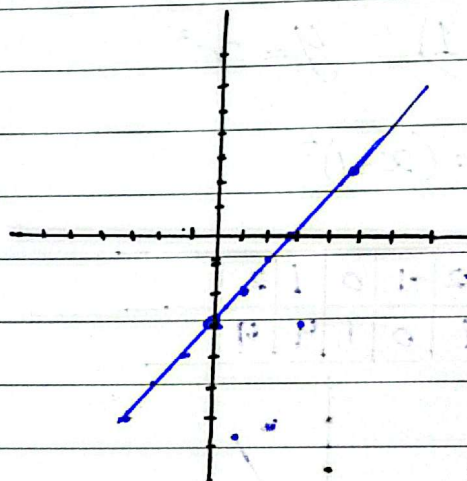
a) $y = x + 3$



x	-3	-2	-1	0	1	2	3
y	0	1	2	3	4	5	6

upward by 3 unit

b) $y = x - 3$



x	-3	-2	-1	0	1	2	3
y	-6	-5	-4	-3	-2	-1	0

Downward by 3 unit

∴ Combine result of shift
(vertical shift + horizontal shift)

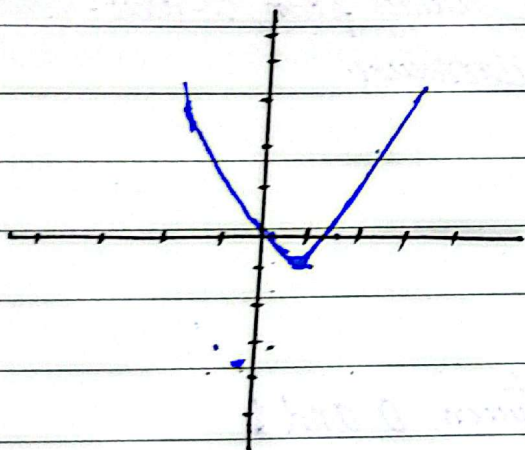
For a function written as:

$$y = f(x - c) + c$$

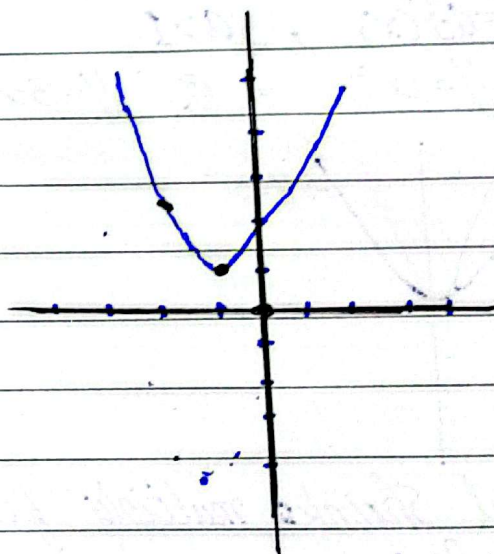
- Inside the bracket $(x - c) \Rightarrow$ Horizontal shift.
 - $\rightarrow (x - c) \Rightarrow$ Shift right side by unit c .
 - $\rightarrow (x + c) \Rightarrow$ Shift left side by unit c .
- Outside of bracket $+c \Rightarrow$ Vertical Shift.
 - $\rightarrow +c \Rightarrow$ move upward
 - $\rightarrow -c \Rightarrow$ move downward.

Applying the conditions.

1) $y = (x - 2)^2 - 1$



2) $y = (x + 2)^2 + 1$



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∴ Two types of Transformation

i) Rigid Transformation:

Horizontal shifts, vertical shifts and reflections are rigid transformation because the basic shape of graph is unchanged.

- These transformations changes only the position of the graph in coordinate plane.

ii) Nonrigid transforms:

They cause a distortion - a change in shape of original graph.

@ For example:

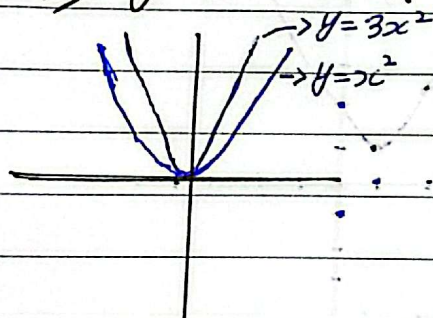
A non rigid transformation of the graph of $y=f(x)$ is represented by $g(x)=cf(x)$, where the transformation is a vertical stretch, when $c>1$.

Vertical Stretch and vertical Shrink.

- ~~Vertical~~ Vertical stretch: multiply $f(x)$ by a number greater than 1

$$y=af(x), \quad a>1$$

1) $y=x^2 \Rightarrow @ \quad y=3x^2$

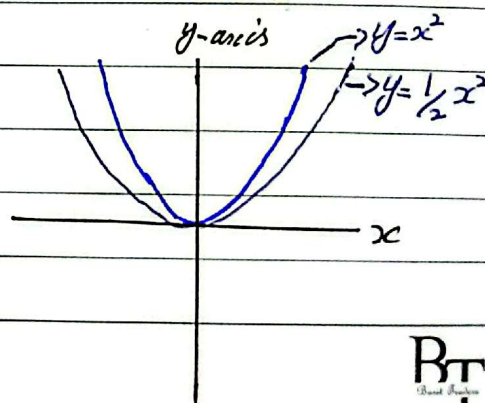


- Stretches the graph away from x-axis.
- Makes the graph narrower.

- Vertical Shrink: multiply $f(x)$ by a number between 0 and 1

$$y=af(x) \quad 0 < a < 1$$

1) $y=x^2 \Rightarrow y=\frac{1}{2}x^2$



- Shrinks the graph towards x-axis.
- Make the graph wider.